

Kazi Sohrab Uddin Titu

AI & ML Engineer · Agentic AI · LLM Fine-Tuning · RAG · LLMOps

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PROFESSIONAL SUMMARY

AI & ML Engineer with hands-on production experience in agentic LLM systems, context engineering, RAG pipelines, and LLM fine-tuning. Proven track record shipping evaluated, monitored AI features using OpenAI, Anthropic (Claude), Gemini, and Grok APIs. Experienced in QLoRA / LoRA fine-tuning of language models (Qwen3, Gemma), LLMOps with Docker and AWS, and building structured evaluation frameworks with measurable outcomes. Strong believer in hypothesis-driven experimentation and real-world impact over demos.

WORK EXPERIENCE

Artificial Intelligence Engineer — Oct 2025 – Present

NET7 · Pisa, Italy

- Designed and deployed agentic LLM systems integrating OpenAI, Anthropic (Claude), Gemini, and Grok APIs for production conversational applications.
- Built scalable RAG pipelines using Qdrant and Elasticsearch — improving factual grounding and reducing hallucination rates through structured retrieval strategies, chunking design, and prompt engineering.
- Defined evaluation frameworks with measurable success metrics (accuracy, latency, relevance, regression detection) to benchmark and iterate on LLM performance.
- Designed n8n-based AI workflow orchestration for multi-agent routing, Elasticsearch integration, and automated quality monitoring.
- Built a dual-interface MCP Server + FastAPI REST API for graph-based component connectivity analysis (piping/wiring systems), featuring BFS path finding and neighbor discovery; deployed via Docker on Render.
- Led end-to-end Text-to-SQL fine-tuning pipeline for Valmet (manufacturing analytics): built a synthetic data generation system with multi-model benchmarking, fine-tuned Qwen3 (0.6B–4B) with QLoRA, and achieved 83.7% eval pass rate on live PostgreSQL queries.
- Deployed containerised AI services with Docker on AWS; leveraged Azure ML for experimentation and model evaluation workflows.
- Contributed to internal knowledge sharing on LLM evaluation best practices and agentic system design across backend, frontend, and product teams.

Trainee Artificial Intelligence Engineer — May 2023 – Aug 2023

ADN Diginet · Dhaka, Bangladesh

- Supported ML experiments through data preprocessing, model training, and evaluation.
- Deployed experimental ML solutions into small-scale application workflows using supervised learning.

KEY PROJECTS

Text-to-SQL Fine-Tuning for Manufacturing Analytics · Valmet / EUROVAST

- Fine-tuned Qwen3 (0.6B, 1.7B, 4B) for natural language to SQL on industrial manufacturing data (tissue paper production, OEE/shift metrics) using QLoRA 4-bit quantization with Unsloth + TRL.
- Ran systematic scaling experiments across 3 model sizes on 141 eval samples: 0.6B → 51.8%, 1.7B → 67.4%, 4B → 83.7% (best). Confirmed that chain-of-thought reasoning (<think> blocks) outperforms SQL-only training by +4.3%.
- Best model (Qwen3-4B QLoRA, LoRA rank 64, 3 epochs): 83.7% eval pass rate on live PostgreSQL queries covering OEE metrics, multi-table JOINS, CTEs, and JSONB manufacturing data.
- Built a modular synthetic data pipeline generating 1,000 labeled examples across 6 personas × 6 intent types × 3 difficulty levels, with 2-layer validation (auto + LLM scoring) and live DB execution checks.
- Benchmarked 4 LLM generators (Claude Haiku, Claude Sonnet, GPT-4.1, GPT-4.1-mini) for training data quality; implemented GRPO reward-based training.

Technologies: Unsloth, TRL, QLoRA, vLLM, HuggingFace, Docker, PostgreSQL, Anthropic API, OpenAI API

AI Assistant · Agentic Conversational AI + RAG Knowledge System

- Built a production-grade agentic AI assistant for a marine services company providing instant technical support for water-system pipelines.
- Designed agentic architecture combining LLM reasoning with structured retrieval; built OCR, document ingestion, and embedding pipelines for large technical manuals.
- Implemented structured evaluation measuring response relevance and consistency; deployed via Docker with Qdrant vector search and API-based LLM orchestration.

Technologies: LLM APIs, RAG, Agentic AI, Qdrant, Elasticsearch, Docker

ATS Graph MCP Server · Dual-Interface Graph Analysis System

- Built a dual-interface system providing both an MCP Server (Model Context Protocol) and FastAPI REST API for piping/wiring component connectivity analysis.
- Implemented BFS-based path finding, neighbor discovery, and flexible component resolution supporting multiple ID formats.
- Deployed via Docker on Render with nginx configuration; demonstrates end-to-end MCP server design for AI assistant integration.

Technologies: MCP / FastMCP, FastAPI, Python, Docker, Render, BFS graph algorithms

YOLOv8 Fine-Tuning & Serverless Deployment · Modal + FastAPI

- Fine-tuned YOLOv8 for custom object detection and deployed as a scalable serverless inference API on Modal using FastAPI — demonstrating end-to-end MLOps from training to cloud production.

Technologies: YOLOv8, Modal, FastAPI, Python · github.com/kazisohrabuddintitu/Modal_FastApi

Cattle Weight Prediction · Computer Vision + Supervised ML

- Trained and evaluated computer vision + supervised ML models to predict cattle weight from a single side-profile image; integrated inference pipeline into a Django application for real-time predictions.

Technologies: Computer Vision, Supervised ML, Django, Flutter

EDUCATION

MSc Computer Science — Artificial Intelligence · Sept 2025 – Present

University of Pisa · Italy

MSc Computer Science — Machine Learning · Sept 2024 – Feb 2025

University of Copenhagen · Denmark

BSc Computer Science & Engineering (CGPA 3.92 / 4.00) · Graduated 2023

Independent University, Bangladesh

TECHNICAL SKILLS

LLM & AI	Agentic AI · LLM Fine-Tuning (QLoRA / LoRA / SFT / GRPO) · Synthetic Data Generation · Context Engineering · RAG · Prompt Engineering · LLM Evaluation · NLP · Reinforcement Learning
LLM APIs & Frameworks	OpenAI API · Anthropic API (Claude) · Google Gemini API · Grok API · LangChain · MCP / FastMCP · Unsloth · TRL · vLLM · HuggingFace
ML / DL	PyTorch · TensorFlow · scikit-learn · YOLOv8 · Supervised & Unsupervised Learning · Neural Networks
LLMOps / MLOps	Docker · AWS · Azure ML · Modal · n8n (workflow orchestration) · Model evaluation & benchmarking · Monitoring
Vector & Search	Qdrant · Elasticsearch
Backend & Web	Python · FastAPI · Django · JavaScript · Java · C++ · REST APIs · PostgreSQL · MySQL · Vue.js
Tools	Git · Colab · HuggingFace Hub · Tailwind CSS

CERTIFICATIONS

- Intermediate Machine Learning — Kaggle · May 2023
- Intro to Deep Learning — Kaggle · May 2023
- Intro to Machine Learning — Kaggle · May 2023
- JavaScript Algorithms and Data Structures — freeCodeCamp · Nov 2022
- Responsive Web Design — freeCodeCamp · Nov 2022